

OOPS-2 →

⇒ { continue Constructors
deep / shallow -
static keyword
Inheritance -

constructors

↳

① Default constructor :- Java ,

X ② Student () {

=

}

③ → Student (String name) {

=

}

④ → Student (String name, int age) {

=

}

3 & 4
- available

new Student();

?

↓
error

copy constructor

```
s1 = new Student();
```

```
s1.age = 40;
```

```
s1.psp = 90.0;
```

```
s1.name = "Mohit";
```

↑
create a copy of this student obj!

~~Student s2 = s1;~~ } copy of ref not the st_obj

@4000
↓
Student s2 = new Student();
s1 @3000
@3000 ~~s2 = s1;~~ X

✓
s2.age = s1.age;
s2.psp = s1.psp;
s2.name = s1.name;

```
Student (Student other) {  
    this.age = other.age;  
    this.psp = other.psp;  
    ↑  
Student st2 = new Student(st1);  
    ↓
```

```

Student {
    int id;
    Exam exam;
}

```

```

Student () {
    =
}

```

```

Student (Student other) {

```

```

    this.id = other.id;

```

```

    this.exam = other.exam;

```

```

}

```

```

}

```

```

    this.exam = new Exam();

```

copying the ref

Deep Copy

```

Student st1 = new Student();

```

```

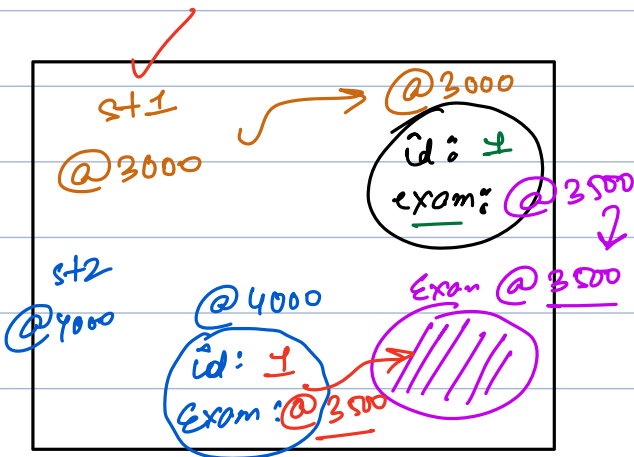
st1.id = 1;

```

```

st1.exam = new Exam();

```



shallow copy

```

Student st2 = new Student(st1);

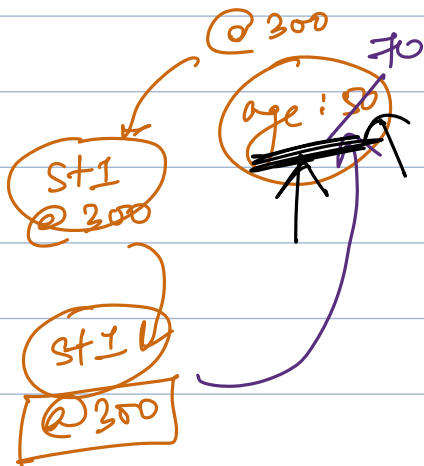
```

```

fun( student st1 ) {
    st1.age = 20;
}

```

}



```

Student st1 = new Student();

```

```

st1.age = 50;

```

```

fun( st1 );

```

```

print( st1.age );

```

70 50

```

fun( int x ) {
    x += 10;
}

```

```

int x = 10;

```

```

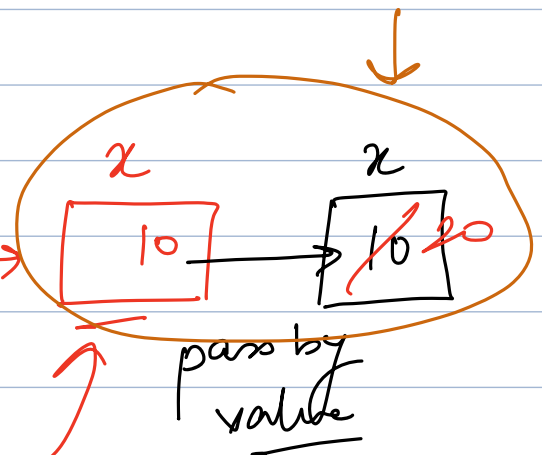
fun( x );

```

```

print( x );

```



10:20 pm

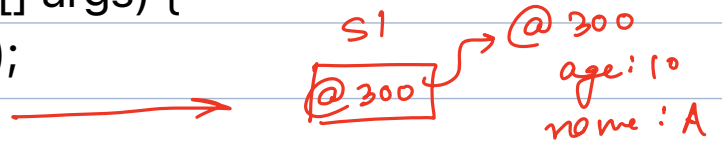
```
public class Client {
```

```
    public static void main(String[] args) {
```

```
        Student s1 = new Student();
```

```
        s1.age = 10;
```

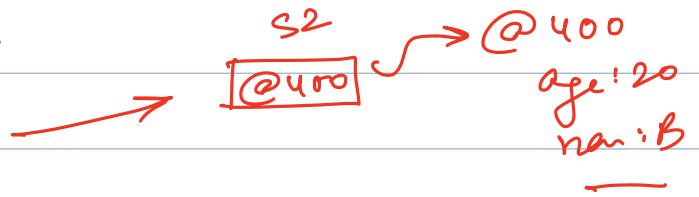
```
        s1.name = "A";
```



```
        Student s2 = new Student();
```

```
        s2.age = 20;
```

```
        s2.name = "B";
```

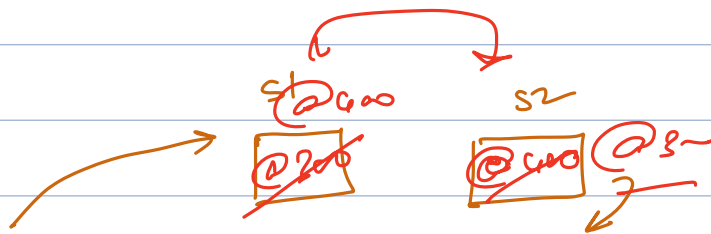


```
        swap(s1, s2);
```

@300, @400

```
        s1.display();
```

```
    }
```



```
private static void swap(Student s1, Student s2) {
```

```
    Student temp = s1;
```

```
    s1 = s2;
```

```
    s2 = temp;
```

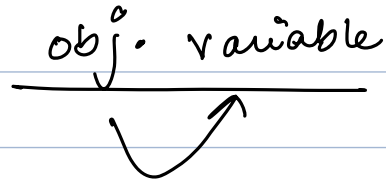
```
}
```

```
}
```

*making change in
ref variable
— scope — swap —*

Static

obj. variable



age ← Student st1 = new Student()

age ← student st2 = new Student();

age — instance variable

Class variables — Class

